RES-5 and the ordered-BCC precedent: the leading selection is

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The operator raised a decisive devil's-advocate point: the reason TECT selected Reading-H is that the ordered BCC FAILED – and RES-5 lives in the same strong-fluctuation regime, so is the screening route not the same kind of over-optimism that failed the ordered BCC? This note addresses it: it identifies the one decisive difference (the leading selection is non-perturbatively proven), grants the parallel's force at higher loops, and accordingly reduces the screening route to necessary-but-not-sufficient. Scope: the reliability question; anchor μ^2.

2. The ordered-BCC failure

The ordered-BCC reading was a MEAN-FIELD prediction (BCC ordered ground state). The LEADING (one-loop) fluctuation correction OVERTURNED it – the Brazovskii restoration to disorder – an $O(1)$ effect that FLIPPED the conclusion (NG-2026-legacy-ordered-vacuum). Mean field was qualitatively wrong.

3. The decisive difference: the leading selection is non-perturbative

The Reading-H leading selection is NOT a mean-field prediction. It is the one-loop result itself, and it is proven NON-PERTURBATIVELY by operator monotonicity (R-U10-3 / neargap-common-mode-resolution): at fixed I the dressing $D_0(I) = \hat{r}$ is pattern-independent, and

$$\Delta F^{(2)} = \tfrac{1}{2}\,\mathrm{Tr}\big[\ln(D_0 + \lambda'\mathcal{P}^2) - \ln D_0\big] \;\geq\; 0 \quad \text{for ANY competitor } \mathcal{P},$$

since $\lambda'\mathcal{P}^2 \succeq 0$ and ln is operator-monotone (verified pointwise, min $\frac{1}{2}[\ln(D_0 + \lambda'\mathcal{P}^2) - \ln D_0] = +4.7 \times 10^{-5} > 0$). Thus $\mathcal{R}_H$ IS the fluctuation-restored state, and the strong fluctuations live in the COMMON sea $D_0(I)$ and CANCEL at leading order. The ordered-BCC error – a leading fluctuation flipping a mean-field sign – is structurally NOT repeated.

4. The parallel's force, granted

The operator is right where it matters: $g = \lambda' B_d = 1.03 \sim 1$ is the SAME strong-fluctuation regime. At HIGHER loops (RES-5) my screening / RPA argument is PERTURBATIVE, and the ordered-BCC lesson is precisely that a perturbative bound is not to be trusted at strong coupling. The screening factor $1/(1+g)$ shows the resummation is FINITE (necessary), but it does NOT establish that the higher-loop corrections preserve the common-mode cancellation (sufficient). My screening note's confidence is hereby reduced.

5. The sharpened residual

$$\boxed{\text{RES-5 residual} = \text{non-perturbative higher-loop common-mode cancellation.}}$$

That is: does the operator-monotonicity-type protection of $\Delta F^{(2)} \geq 0$ extend to the higher-loop / 2PI level, so that the strong common fluctuations still cancel in the difference and only the a_0-suppressed pattern-dependent part survives? The screening route is NECESSARY (finite resummation) but NOT SUFFICIENT; the certificate must VERIFY this higher-loop common-mode cancellation (a sign / monotonicity structure at 2PI order), not assume it.

6. Status

T3 / DEVIL'S-ADVOCATE. Established: the leading selection is non-perturbatively protected (operator monotonicity), so RES-5 is not the ordered-BCC error at leading order; but the strong-coupling regime makes the screening route contingent. **OPEN (sharpened):** the non-perturbative higher-loop common-mode cancellation. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. The screening note (res5-2pi-resummation-strategy) is annotated: NECESSARY-but-not-sufficient, contingent on this residual.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"So the screening route was over-claimed."* PARTLY-UPHELD: the route is real (the resummation IS finite, repulsive), but its sufficiency was over-stated; this note reduces it to necessary-but-not-sufficient and names the higher-loop common-mode cancellation as the load-bearing residual. The operator's catch is recorded as a confidence reduction.

(β) *"Then is Reading-H itself unsafe, like the ordered BCC?"* DISMISSED at leading order: $\Delta F^{(2)} \geq 0$ is non-perturbative (operator monotonicity), so the LEADING selection is safe regardless of fluctuation strength. The risk is confined to whether higher loops overturn a non-perturbatively-correct leading result – a strictly weaker worry than the ordered-BCC mean-field flip.

(γ) *"Does the strong coupling threaten the operator- monotonicity proof?"* DISMISSED: operator monotonicity of ln is exact for any PSD $\lambda'\mathcal{P}^2$, independent of the coupling strength; it is an algebraic inequality, not a perturbative estimate.

Quantitative sanity check (mandatory): *non-perturbative* – $\min \frac{1}{2}[\ln(D_0+\lambda'\mathcal{P}^2)-\ln D_0] = +4.7\times 10^{-5} > 0$ (operator monotonicity, coupling-independent); *regime* – $g = 1.03 \sim 1$ (same as ordered-BCC failure); *structure* – strong fluctuations in the common $D_0(I)$ (cancel at leading order); *logic* – leading non-perturbative + higher-loop contingent.

8. Result footer

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Result ID:              B1-RH-ENUM / RES-5 ordered-BCC parallel (reliability)
Precise statement:      RES-5 is in the same strong-fluctuation regime (g=lam' B_d
                        =1.03~1) that failed the ordered BCC. DIFFERENCE: the
                        Reading-H leading selection dF^(2)=(1/2)Tr[ln(D_0+lam'P^2)
                        -ln D_0]>=0 is NON-PERTURBATIVE (operator monotonicity,
                        R-U10-3; min +4.7e-5>0), so R_H is the restored state, not
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	a mean field -- the ordered-BCC error is not repeated at leading order. But at higher loops the screening/RPA argument is perturbative => CONTINGENT (the ordered-BCC lesson). Load-bearing residual: the NON-PERTURBATIVE higher-loop common-mode cancellation; the screening route is necessary-but-not-sufficient.
Scope:	Reliability / devil's-advocate; anchor mu^2.
Dependencies:	res5-2pi-resummation-strategy (downgraded here); neargap-common-mode-resolution (R-U10-3, operator monotonicity); NG-2026-legacy-ordered-vacuum; Math424-AddA.
Evidence grade:	ANALYTIC + EXECUTED (res5_orderedbcc_parallel.py 4/4).
Reproduction command:	python codes/vacuum/res5_orderedbcc_parallel.py
Expected output:	dF^(2)>=0 pointwise (min +4.7e-5); g=1.03; 4/4 PASS.
Falsification gate:	A competitor P with dF^(2)<0 (would break operator monotonicity -- impossible for PSD lam'P^2); or a higher-loop pattern-dependent O(1) term surviving the common-mode cancellation (would make RES-5 the ordered-BCC error at higher loops).
Tier before / after:	No tier flip. B1 T6 on {H-LAYER}. Screening route reduced to necessary-but-not-sufficient; residual sharpened to the non-perturbative higher-loop common-mode cancellation.
No-overclaim statement:	This does NOT close RES-5 and reduces the screening route's claim. The leading selection is non-perturbatively safe; the higher-loop reliability is contingent on the non-perturbative common-mode cancellation, which the certificate must VERIFY (not assume). The operator's ordered-BCC parallel is upheld as a strong-coupling caution.
Next required action:	The 2PI certificate must establish the higher-loop common-mode cancellation via a sign / operator-monotonicity structure at 2PI order (extending R-U10-3), BEFORE the screened quantitative bound is trusted.